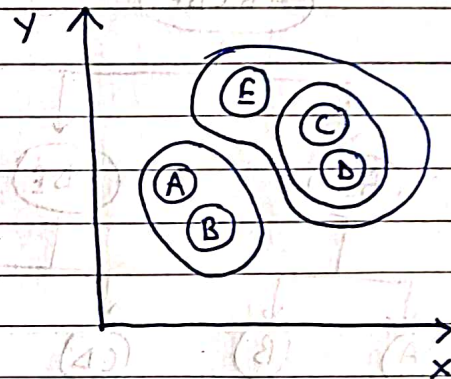
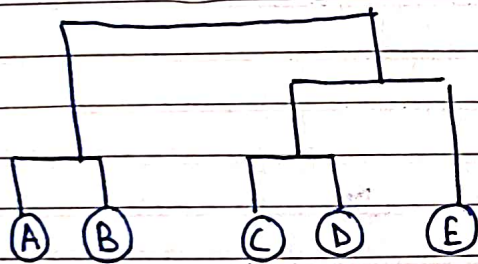


* Hierarchical cluster

Step by step we build cluster based on nearest neighbour.

2 parts $\left\{ \begin{array}{l} \text{agglomerative clustering} \\ \text{divisive clustering} \end{array} \right.$

① Agglomerative clustering



* The lines drawn to connect the points are called dendrograms.

* C is nearest to D hence they are connected first. Next A is nearest to B hence they are connected. Next E is nearest to CD cluster hence E is connected to CD. Next AB and CDE are connected.

* In agglomerative cluster with the help of dendrogram we build the cluster from bottom to top approach.

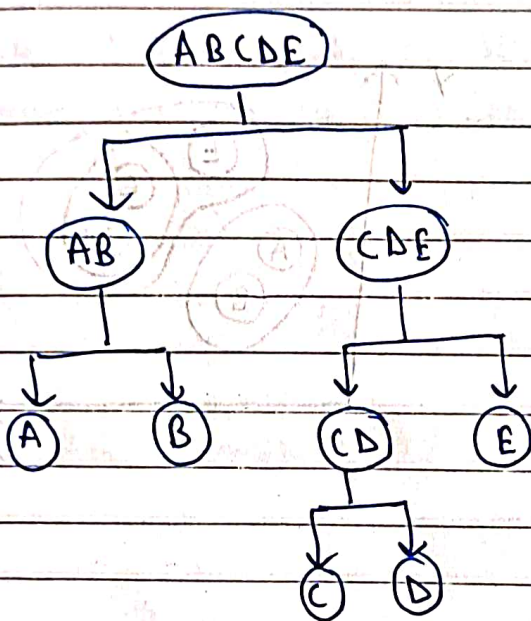
* We build the cluster using bottom to top approach based on nearest neighbour.

(2) divisive cluster

divide the cluster

ABCDE $\rightarrow$ AB and CDE $\rightarrow$ AB and CD, E $\rightarrow$ A, B and CD, E
 $\rightarrow$ A, B and C, D, E

From cluster to individual divided points



* We divide the cluster to individual data points top to bottom approach.

* Difference b/w KMeans vs Hierarchical
In hierarchical we don't have centroid.